

Tax Productivity and Development: A Quantitative Macroeconomic Analysis

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Abstract

This paper measures productivity in tax collection using new cross-country evidence from 98 countries spanning many years and a wide range of income levels. Our main measure of tax productivity is the ratio of aggregate taxes collected to tax administrative expenses. We document that tax productivity is strongly increasing in GDP per capita, both in the cross-section and over time, with the richest countries collecting about ten times as many tax revenues per dollar spent as the poorest. We interpret this finding in a Ramsey model in which governments choose effective tax rates, taking as given their productivity levels in taxation and producing output. Private and public capital are complements in the production function as in Barro (1990) and Baxter-King (1993), and public capital can only be financed through taxation. We use our model to simulate the long-run aggregate effects of raising tax productivity in the developing world, finding significant quantitative effects.

Keywords: tax productivity, state capacity, Ramsey problem, macroeconomic development

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1 Introduction

One of the most enticing theories of long-run economic development places state capacity — and in particular the ability of states to raise tax revenue — at the heart of the development process (see e.g. [Besley and Persson, 2011](#)). Empirical work by [Mayshar, Moav, and Pascali \(2022\)](#) shows that the most successful early states, such as Egypt, were those who were better able to raise revenues that they used to pay for public works and military ventures. [Tilly \(1990\)](#), [Dincecco and Katz \(2016\)](#), and [Gennaioli and Voth \(2015\)](#) provide evidence that the constant wars waged by European states in the 16th and 17th centuries led them to build up state capacity, and may ultimately have sown the seeds of their future development. Recent work by [D’Arcy, Nistotskaya, and Olsson \(2024\)](#) studying tax systems throughout recorded history ties the creation of public records of land ownership, which form the basis of property taxation, to long-run growth.

The link between a government’s ability to collect taxes and its long-run level of development has received less attention in the macroeconomic literature on growth and development. The foundational papers of [Barro \(1990\)](#), [Baxter and King \(1993\)](#), and [Jones, Manuelli, and Rossi \(1993\)](#) incorporate public capital stocks into the production function, but abstract from any costs of collecting the taxes needed to finance public investments. Models of optimal taxation of infinitely lived agents ([Judd, 1985](#); [Chamley, 1986](#)), overlapping generations ([Conesa, Kitao, and Krueger, 2009](#)), or Aiyagari-type heterogeneous agents ([Auclert, Cai, Rognlie, and Straub, 2024](#)), similarly abstract from the resources needed to mobilize tax revenues, treating tax rates as freely chosen by governments. A notable exception is the paper by [Besley, Ilzetzki, and Persson \(2013\)](#), which incorporates the political economy of tax capacity into a growth model, finding qualitatively that more severe fiscal constraints can lead an economy to reach a less desirable steady state.

The goal of this paper is to take a quantitative macroeconomic approach to understanding the role of tax capacity in the development process. We focus on what we call tax productivity, which represents the amount of resources a government needs to use up in order to collect taxes. We first measure tax productivity in a large set of countries at all levels of development. We then use these data to help build and calibrate a growth model with public and private capital in the production function as in [Baxter and King \(1993\)](#), in which a government chooses its optimal tax rate, as in the literature on Ramsey taxation. The key novel feature of the model is a tax administration sector that can collect tax revenues from households but only by using up inputs of labor (which can be thought of as tax collectors) in the process. Countries differ exogenously in their tax productivity, or efficiency of turning inputs into tax revenues. We use the model to simulate the long-run aggregate effects of raising tax productivity for countries of low income levels, where we suspect growth is most constrained by an inability to mobilize tax revenues.

We begin by drawing on new evidence that we hand-collected from tax administrations, or tax authorities, from 98 countries around the world and covering 625 country-year observations. Examples of tax authorities are the U.S. Internal Revenue Service and Ghana's Revenue Authority. We measure tax productivity using data on aggregate tax revenues and the administrative cost of collecting those revenues. Our main measure of tax productivity is simply the ratio of aggregate tax revenues to collection costs, and represents the average amount collected per dollar spent collecting. This unitless productivity measure is easy to interpret and compare across countries and over time within countries.

Our main empirical finding is that tax productivity is strongly increasing and GDP per capita. In terms of magnitudes, the poorest countries in the world collect about \$30 for every dollar they spend collecting taxes; this number rises to \$300 in the world's richest countries. We show that this pattern is not an artifact of differences in scope of the revenue authorities' activities. For example, the strongly increasing tax productivity with development is still present when excluding tax administrations that are not responsible for VAT/sales tax or customs duties, or those that *are* responsible for lotteries or social security transfers.

Exploiting the panel nature of our data, we ask how tax productivity changes with GDP per capita within countries, using regressions with country-fixed effects. We find that countries tend to raise their tax productivity as they grow, and the elasticity of tax productivity to GDP per capita is even stronger than in the cross-section. Our longest time series comes from the United States, where we measure tax productivity back from 1867 to the present. Interestingly, the U.S. tax productivity around the time of the Civil War was comparable to that of developing countries today. It then rose rapidly until around World War II, with slower growth since. As in the cross-section, the pattern of rising tax productivity with development holds just as well, and looking only at tax administrations that have a common set of duties.

We interpret these facts in a Ramsey model that features a tax administration sector that can raise tax revenues from households but using inputs of labor up in the process. The government chooses tax collection efforts in order to maximize the utility of an infinitely lived representative household. Production is neoclassical and turns inputs of labor, private capital, and public capital into a single output good that can be used for either consumption or investment. Public capital can only be provided through tax revenue, and taxes can only be raised through costly efforts by the tax administration. The government takes its tax productivity as well as its productivity in making final goods as given. The household has a continuum of members that it can deploy into either a modern sector, that uses both types of capital but faces taxation of income, or a (less productive) traditional sector that can avoid taxation entirely. Labor can also be used as an input into the tax administration sector.

We analytically characterize the model, focusing primarily on its steady state. The optimal tax collection rule highlights several key trade-offs. When private capital yields high returns, the government keeps resources within firms to allocate toward private capital rather than increasing public spending on tax collection. Conversely, when public capital significantly increases firm productivity, investing in public capital becomes more attractive. Additionally, improvements in the efficiency of tax collection boost the effectiveness of tax collectors, allowing the government to generate more revenue. Lastly, an expansion of the modern sector increases the overall tax base, further enhancing total tax collections.

In quantitative work (in progress), we simulate the effects of raising tax productivity on aggregate outcomes like GDP per capita, and stocks of public and private capital. We generally find strong long-run responses of output to tax productivity. In future work we plan to discipline our quantitative analyses using our cross-country evidence as well as other estimates of the marginal cost of collecting taxes in low-income countries.

Our paper contributes to the macroeconomic literature on taxation by analyzing the effects of a resource cost of collecting revenues. [Straub and Werning \(2020\)](#) overturn the famous result, inspired by Chamley and Judd, that in Ramsey models the optimal capital tax rate should be zero. We take a simple route and assume only taxation of labor, and positive tax rates are always optimal in our setting since public capital is complementary to private capital. Our study also abstracts from insurance considerations as in Mirrlees problems ([Kapicka, 2013](#); [Farhi and Werning, 2013](#); [Golosov, Troshkin, and Tsyvinski, 2016](#); [Boerma and McGrattan, 2020](#)). Our paper is related to the study by [Bhandari, Evans, Yao, and McGrattan \(2024\)](#) incorporates tax evasion into a macro model calibrated to match U.S. evidence on the effects of tax audits. More generally, our work relates to the public finance literature on tax evasion (see [Slemrod, 2007](#), and the references therein). Households in our model can evade taxes by working in the traditional sector, though only by accepting a drop in productivity. Similarly, Governments can raise effective tax rates in our setup by spending more resources on tax administration, which is supposed to represent greater effects to enforce the tax code.

Finally, our paper adds to the large literature on public finance and development focused on tax capacity and its determinants ([Gordon and Li, 2009](#)). Paths to increasing tax capacity emphasized by this literature include incentive pay for tax collectors ([Khan, Khwaja, and Olken, 2015](#)), collaboration with traditional leaders, ([Balan, Bergeron, Tourek, and Weigel, 2022](#)), information technology ([Hjort and Tian, 2024](#); [Okunogbe and Tourek, 2024](#)), including using geo-referenced databases of taxpayers ([Dzansi, Jensen, Lagakos, and Telli, 2025](#)), and self-enforcing tax systems ([Naritomi, 2019](#)). Much of the recent literature has focused on taxation of properties, which holds vast promise but is thought to be vastly under-collected in the developing world ([Brockmeyer, Es-](#)

tefan, Arras, and Suárez Serrato, 2023). The same is true of taxes on labor income, particularly given the challenges in collecting from the self-employed (Jensen, 2022). Our study contributes by modeling and quantifying the long-run general equilibrium consequences of improving tax capacity.

2 Data and Empirical Findings

We begin by presenting the data we use in our analysis, and defining our measure of tax productivity. We then summarize cross-country and time series patterns (for the United States) on how tax productivity varies with income per capita.

2.1 Data

In this subsection, we describe the data collection exercise undertaken to measure our proxy of government tax productivity in a large sample of countries and years. Specifically, our proxy is the ratio of taxes collected to expenditure on tax collection.

The numerator in this ratio captures taxes collected at the national level. The taxes that are typically collected at this level are on personal and corporate income, property, wealth and inheritance and consumption of goods and services. Importantly, we include data on taxes collected at the border. These choices imply that we are focusing on creating a measure of tax collection for the national government who is in charge of collecting taxes both domestically and at the border. The concept of tax collection productivity can also be applied to the tax authorities at the sub-national levels of government (e.g. state or county). Such an exercise would be an interesting extension; for now, our data collection implies a focus on national tax authorities. We focus on actual taxes collected, rather than projected or estimated taxes. Finally, whenever the data permits, we ensure that non-tax revenues are excluded.

The denominator in this ratio seeks to measure the annual expenditure of the national tax authority. More specifically, we seek to measure the sum of operating expenditures and capital expenditures. In the first category, by far the most important item is salary; other items include training and miscellaneous administrative expenses. The second category includes expenses that go towards purchasing and maintaining buildings and land as well as other types of infrastructure. Expenses in information and communication technology (ICT) can be included in the first or second category, depending on whether they are long-term investments (e.g. purchase of a new server) or day-to-day expenses (e.g. software maintenance). Breakdown of collection expenses by type, which exists on an incomplete basis in our sample, reveals that by far the most important cost-categories are salary and ICT. Whenever the data permits, we try to include as many of these cost-categories as possible: in all countries, we are confident that salaries are included; for some countries, the cost measure will be more comprehensive and include capital expenses.

There are many similarities in tax authorities across countries, and this definition of collection cost

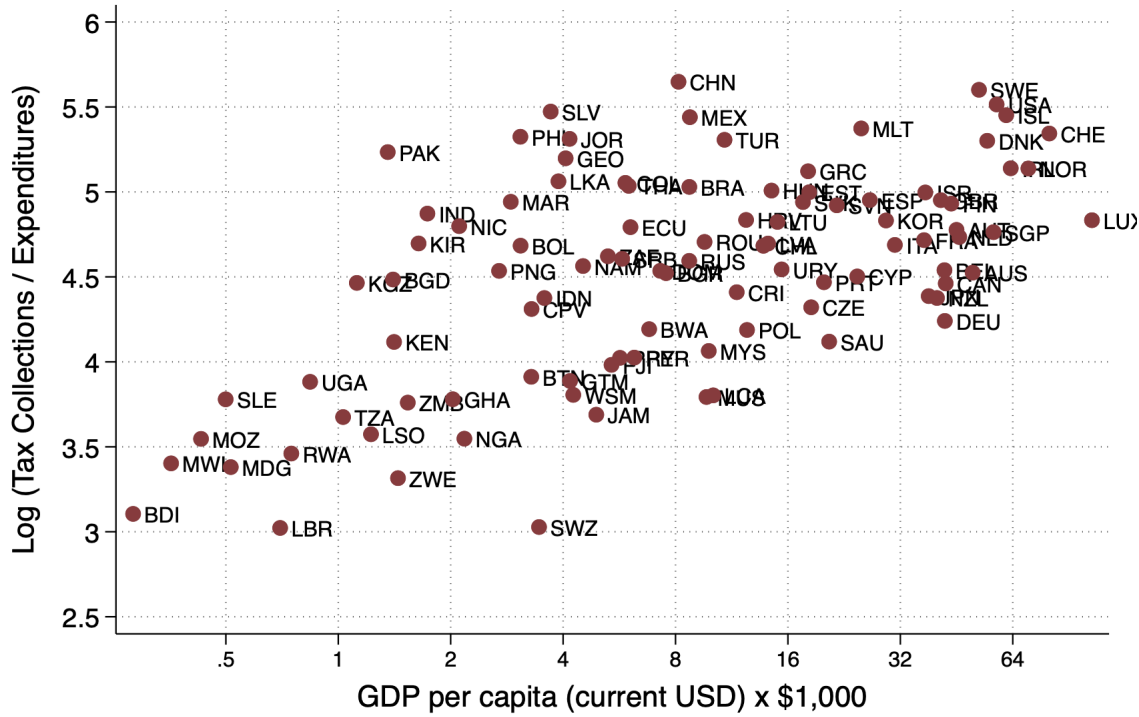
allows us to capture most of them in a manner that is meaningfully comparable across borders and over time. Notwithstanding, there are limits to this measure. First, there are differences across countries in the range and nature of revenues administered. Here, the most significant distinction is probably between taxes administered by the customs authority (tariffs and other border taxes) versus taxes administered by the domestic tax authority. There is variation across countries, and within countries over time, in the roles assigned to each institution; importantly, our measure seeks to capture the combined cost of collecting taxes domestically and at the border – such that differences across space and time in roles assigned to different national institutions should not affect our ratio. Another example is that some countries administer direct taxes at the national level but some indirect taxes at the sub-national level (e.g. the collection of sales taxes is devolved to states in both India and the US). Second, some tax authorities are responsible for carrying out activities that are not related to tax administration (e.g. administering certain welfare benefits or maintaining a national population registry). Third, there are naturally other factors which influence this ratio and which are not related to tax productivity: for example, macroeconomic changes could, at least temporarily, create changes in the taxable base without corresponding adjustments to collection expenditures.

We collect data from several sources in a cross-section of countries. The most important source is the International Survey on Revenue Administration (ISORA), which is administered collaboratively by multiple international agencies including the OECD, the IMF and the Inter-American Center of Tax Administrations. Whenever available, we use data on both the numerator and denominator from this source. We complement this with data from individual countries, based on the yearly government budget and national reports from the tax authority and national statistics office. For each country, we seek to collect multiple years of data between 2012 and 2018; we report the yearly average in that range, to smooth out fluctuations that occur in the ratio due to abnormal levels of expenses or collection in any given year.

Our resulting cross-country sample contains data for 98 countries. The countries span a broad range of development levels, from Burundi, Malawi and Pakistan, to Bolivia, the Philippines and Jordan, to Denmark, Singapore and the United States. We have a total of 625 country-year observations, and for 16 countries we have relatively long time series spanning multiple decades.

Our longest time series comes from the United States, where we can measure tax productivity back to 1867. We rely on annual reports published on a yearly basis by the commissioner of the Internal Revenue Service (IRS). We measure tax collections as gross collections, before refunds are issued, but after including penalties and interest paid towards outstanding taxpayer arrears. Importantly, in contrast to the cross-country data, We include only domestic collections, and exclude amounts collected at the border as part of trade taxes. We include all domestic expenditures

Figure 1: Tax Productivity in the Cross-Section of Countries



on collection, and exclude the cost of collecting taxes at the border. The expenditure measure is comprehensive and includes all expenditures associated with the process of collecting internal revenue. Throughout the entire period studied, personnel costs (including for central officers and field agents) has been an important cost-component. Over time, other cost components have become gradually more important, including physical infrastructure and ICT.

2.2 Results: Tax Productivity and Development

Figure 1 shows the association across countries between the (log) of GDP per capita in USD and the log of the ratio of taxes collected to collection expenses. There is a clear positive association, whereby tax authorities in countries with higher GDP per capita collect more taxes per dollar of tax expenditure. The relationship generally appears to hold within regions of the world, with stronger slopes in Sub-Saharan Africa and Europe and Central Asia, and flatter slopes South Asia and Middle East and North Africa.

The magnitude of the differences in tax productivity between the world's poorest and richest countries are substantial. Countries like Burundi, Liberia, and Swaziland collect roughly \$30 for every dollar they spend collecting taxes. The world highest value come in much richer countries like

Table 1: Country-Level Regressions: Tax Productivity and Income Per Capita

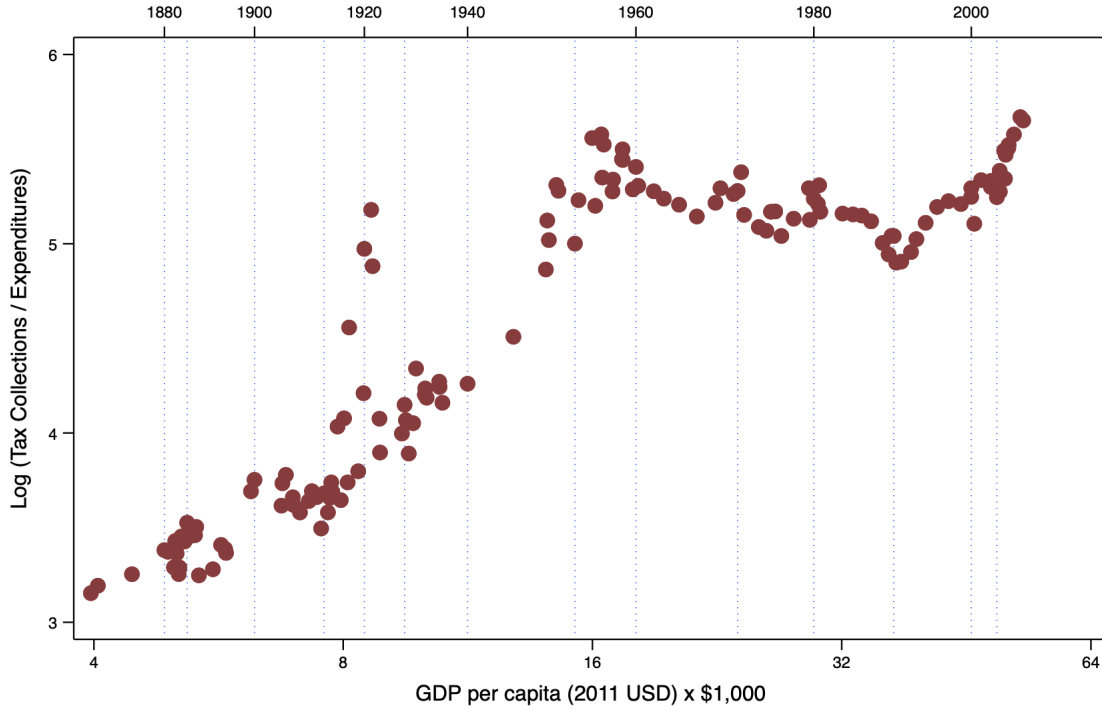
	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for VAT/sales taxes	(3) Log (T/X) Exclude if not responsible for customs duties	(4) Log (T/X) Exclude if responsible for lotteries	(5) Log (T/X) Exclude if responsible for social security contributions
Log GDP pc	0.256*** (0.0363)	0.257*** (0.0367)	0.268*** (0.0367)	0.258*** (0.0362)	0.222*** (0.0443)
Countries	98	95	88	96	72
R-squared	0.341	0.346	0.384	0.351	0.264

Note: Each column represents one regression of log tax productivity, defined as taxes collected divided by tax administration expenditures, on log GDP per capita. *** means significant at 1%, ** means 5%, and * means 10%

the United States and Sweden, which collect around \$300 per dollar spent. China is closer to the middle of the income distribution but also collects around this much.

In the cross-country sample, there are some differences across nations in the functions of the tax administration and the nature of the administered revenues. Table 1 investigates the robustness of the results to these differences. Column (1) reports the slope coefficient (0.256) that corresponds to the full sample in Figure 1. Columns (2) to (5) remove countries from the sample that differ in some specific way. In column (2), we remove countries where the national tax administration is not responsible for collecting the main indirect tax (either sales tax or value-added tax); this leads to the removal of 3 countries. In column (3), we remove countries where the national tax authority is not responsible for collecting customs duties; this leads to the removal of 10 countries. In column (4), we exclude countries where the national tax authority is not responsible for collecting taxes on lotteries; this leads to the removal of 2 countries. Finally, in column (5), we exclude countries that are also responsible for administering social security contributions; this leads to the removal of 26 countries. Across these subsamples, the slope-coefficient on GDP per capita remains remarkably stable, ranging from 0.268 to 0.222. These results suggest that the positive association between development and tax collection productivity is robust to variation in the range of functions undertaken by the tax authority.

Figure 2: Tax Productivity in the United States: 1867 to 2017



We measure collection expenditure in the United States at the Federal level between 1867 and 2017. We include all domestic expenditures on collection, and exclude the cost of collecting taxes at the border. The expenditure measure is comprehensive and includes all expenditures associated with the process of collecting internal revenue. Throughout the entire period studied, personnel costs (including for central officers and field agents) has been an important cost component: they represent just over 90 percent of total costs in 1880, and just over 70 percent in 2017. Over time, other cost components have gradually become more important, including physical infrastructure and information services.

Figure 2 displays the long-run evolution of the cost of (federal, domestic) tax collection in the US. To maintain the consistency with the cross-country figure, the figure plots this cost of collection against the continuous measure of GDP per capita rather than calendar years. Similar to the cross-country figure, there is a clear, strong positive association: at higher levels of GDP per capita, the IRS in the US collects much more tax revenues per dollar of collection expenses. The break in trend corresponds to the immediate post-WWII period; the positive association is much steeper in the period prior to WWII. Interestingly, the range of observed values for our productivity measure is comparable in the US long-run and across countries today - fluctuating in both cases between (log) 3 and (log) 6 values.

Table 2: Panel Evidence: Tax Productivity and Income per Capita

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for customs duties	(3) Log (T/X) Exclude if responsible for social security contributions	(4) Log (T/X) Exclude if changes in responsibilities
Log Real GDP pc	0.637*** (0.175)	0.636*** (0.197)	0.684*** (0.163)	0.644*** (0.193)
Observations	625	546	519	578
R-squared	0.563	0.541	0.574	0.541
Country FE	yes	yes	yes	yes

Note: Each column represents one regression of log tax productivity on on log GDP per capita with country fixed effects. the sample consists of 6 OECD countries, 1 from Latin America, 4 from Asia, and 5 from Sub-Saharan Africa. *** means significant at 1%, ** means 5%, and * means 10%

Table 2 looks at how tax productivity varies with GDP per capita when looking over time within countries, i.e. with a country fixed effect. Here we see an even stronger relationship than in the cross section. Each column in the table explores different sample restrictions, as above, based on the scope of the tax administrations' duties. The basic finding is that these restrictions again play little role in explaining the magnitude of the elasticity of tax productivity to income per head.

3 Model of Tax Productivity and Development

In what follows, we consider an economy in which a continuum of heterogeneous workers chooses between working in a modern or a traditional sector. The government hires tax collection personnel to collect taxes. Tax revenue finances public capital which enters modern production à la [Baxter and King \(1993\)](#).

3.1 Environment

3.1.1 Agents and Technologies

Workers. There is a continuum of workers with heterogeneous ability (or efficiency units of labor) $x \in [\underline{x}, \bar{x}]$ and cumulative distribution $D(x)$. Each worker chooses to supply labor in one of two sectors. In the modern sector, only workers with ability above a cutoff x^* are employed; that is, total efficiency units in the modern sector is $L_M = \mathbb{E}[x|x \geq x^*]$. Workers in the traditional sector have efficiency units equal to one and labor supply in the traditional sector is denoted as $L_T = D(x^*)$.

Modern Sector. Modern firms produce output according to a non-linear technology that features both private and public capital:

$$Y_M = F(A^M, k_G, k, L_M) \quad (1)$$

where A^M is a technology parameter, k_G denotes public capital, k is private capital, and L_M is the effective labor employed in the modern sector. We assume that the production F is of constant returns to scale in private capital and labor (k, L_M) . In each period t , modern firms hire workers at wage $w_{M,t}$ and rent capital at rate r_t . Their profit maximization problem is

$$\max_{k_t, L_{M,t}} F(A^M, k_{G,t}, k_t, L_{M,t}) - w_{M,t} L_{M,t} - r_t k_t. \quad (2)$$

Traditional Sector. Firms in the traditional sector use a linear production function:

$$Y_T = A^T L_T, \quad (3)$$

where A^T is the productivity parameter and L_T is the labor employed. Traditional firms hire workers at a fixed wage $w_T = A^T$.

Tax Collection. In each period t , Taxes T_t are collected on modern sector wages. However, the tax rate $T_t/w_{M,t}L_{M,t}$ depends positively on the tax collection personnel $L_{C,t}$ and a country-specific tax collection productivity A^C . Let $G(A^C, L_{C,t}) \leq 1$ denote the government tax productivity function that determines the rate of modern sector wages collected so that taxes $T_t = G(A^C, L_{C,t})w_{M,t}L_{M,t}$. Without loss of generality, tax collection personnel is paid at the modern sector wage and we assume that the tax productivity function is has decreasing returns to scale in $L_{C,t}$.

3.1.2 Households

The modern and traditional sector workers live in a household that constitutes the representative consumer. The representative household owns all capital and supplies total labor. The representative household maximizes lifetime utility subject to the intertemporal budget constraint,

$$\begin{aligned} \max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t} + L_{T,t}), \\ c_t + k_{t+1} + b_{t+1} \leq (1 + r_t - \delta)k_t + (1 - G(A^C, L_{C,t}))w_{M,t}L_{M,t} + w_T L_{T,t} + R_t b_t. \end{aligned} \quad (4)$$

with initial private capital $k_0 = \bar{k}_0$. In this equation, r_t is the return on private capital, δ is the depreciation rate of private capital, and R_t is the return on bonds. Note that income taxation affects sectoral allocations.

3.1.3 Government and Public Capital Accumulation

The government hires tax collection personnel and collects taxes. The government can borrow to finance its overall budget. That is the budget constraint:

$$w_{M,t}L_{C,t} + R_t b_t + k_{t+1}^G \leq T_t + b_{t+1} + (1 - \delta^G)k_t^G \quad (5)$$

where δ^G is the depreciation rate of public capital. However, the government cannot borrow to finance public capital,¹ that is public capital evolves according to

$$k_{G,t+1} = (1 - \delta^G)k_{G,t} + T_t, \quad (6)$$

¹We can relax this assumption by assuming that the government borrows at a higher rate than the households to finance its budget. Note that in equilibrium, this market structure for government finances implies that the government can borrow short-term to finance inputs in tax collection –here public sector wages— while public capital is the sum of depreciated values of taxes collected. One interpretation for this model of tax collection and development is that there are payment in advance constraints for tax collectors that require debt financing.

Higher tax revenues directly feed into more public capital, which in turn boosts modern production.

Market clearing in the goods market requires that

$$c_t + k_{t+1} + w_{M,t}L_{C,t} + k_{G,t+1} \leq (1 + r_t - \delta)k_t + w_{M,t}L_{M,t} + A^T L_{T,t} + (1 - \delta^G)k_{G,t}. \quad (7)$$

3.1.4 Equilibrium

A competitive equilibrium in this economy consists of allocations $\{c_t, L_{T,t}, L_{M,t}, L_{C,t}, k_t, k_{G,t}, b_t\}$ and prices $\{w_{T,t}, w_{M,t}, r_t, R_t\}$ such that

1. Firm optimality: Modern firms take prices $\{w_{M,t}, r_t\}$ and public capital $k_{G,t}$ as given, solve modern producer's problem (2) and make zero profit in equilibrium. Traditional firms hire labor at wage $w_t = A^T$,
2. Household optimality: representative households take prices $\{w_{T,t}, w_{M,t}, r_t, R_t\}$ and tax collection personnel $L_{C,t}$ for given and solve the consumer problem (4),
3. Budget constraints: government's budget constraint (5) and public capital accumulation equation (6) are satisfied,
4. Market clearing: markets clear (7).

3.2 The Planner's Problem

To study optimal policy, we now formulate the Ramsey problem. The planner (or tax authority) chooses sequences of tax collection personnel $\pi = \{L_{C,t}\}$ to maximize the representative household's lifetime welfare $\sum_{t=0}^{\infty} \beta^t u(c_t(\pi), L_{M,t}(\pi) + L_{T,t}(\pi))$ where $c_t(\pi), L_{M,t}(\pi), L_{T,t}(\pi)$ are equilibrium allocations with tax collection.²

Following the primal approach, we formulate the planner's problem as a choice on allocations directly. Conditions for these allocations to correspond to those chosen by the households in equilibrium result in an implementability condition that constrains the planner's choices.

Lemma 1. *The implementability condition on the planner's allocation choice is:*

$$\sum_{t=0}^{\infty} \beta^t \left[u_c(t)c_t - u_{L_M}(t)L_{M,t} - u_{L_{T,t}}(t)L_{T,t} \right] = u_c(0)R_0k_0. \quad (8)$$

²In this environment, many tax instruments are equivalent. An increasing tax on consumption is equivalent to taxes on private capital. In addition, we can show similarly to [Chamley \(1986\)](#) that optimal private capital taxes are zero starting in period one. Therefore, we focus on income taxation and the productivity of tax collection.

Proof. The household maximizes

$$\max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t} + L_{T,t})$$

subject to the sequence of budget constraints

$$c_t + k_{t+1} + b_{t+1} = (1 + r_t - \delta)k_t + (1 - G(A^C, L_{C,t}))w_{M,t}L_{M,t} + w_T L_{T,t} + R_t b_t,$$

with given k_0 . Let λ_t be the Lagrange multiplier on the period- t constraint. The corresponding first-order conditions (FOCs) include: For consumption:

$$u_c(c_t, L_t) = \lambda_t.$$

For capital:

$$\lambda_t = \beta \lambda_{t+1} (1 + r_{t+1} - \delta),$$

For bonds:

$$\lambda_t = \beta \lambda_{t+1} R_{t+1}.$$

For labor in each sector: The intratemporal conditions (from differentiating with respect to L_M and L_T) yield

$$u_{L_M}(c_t, L_t) + \lambda_t (1 - G(A^C, L_{C,t})) w_{M,t} = 0,$$

$$u_{L_T}(c_t, L_t) + \lambda_t w_T = 0.$$

where $L_t = L_M + L_T$ is total labor. Multiply the period- t budget constraint by the discount factor and the multiplier:

$$\beta^t \lambda_t [c_t + k_{t+1} + b_{t+1}] = \beta^t \lambda_t [(1 + r_t - \delta)k_t + (1 - G(A^C, L_{C,t}))w_{M,t}L_{M,t} + w_T L_{T,t} + R_t b_t].$$

Now we sum over $t = 0$ to ∞ . Because the asset evolution terms (for k_{t+1} and b_{t+1}) and the Euler equations imply that the multipliers λ_t evolve consistently with the returns, the terms involving future assets cancel out telescopically. In other words, when you write the sequence $\beta^t \lambda_t k_{t+1}$ and $\beta^t \lambda_t b_{t+1}$, the Euler equations ensure that almost all terms cancel with the corresponding terms coming from $\beta^{t+1} \lambda_{t+1} (1 + r_{t+1} - \delta) k_t$ and $\beta^{t+1} \lambda_{t+1} R_{t+1} b_t$. The remaining term is the one associated with the initial capital $\lambda_0 R_0 k_0$. Rearrange the labor FOCs as follows:

$$\lambda_t (1 - G(A^C, L_{C,t})) w_{M,t} = -u_{L_M}(c_t, L_t),$$

$$\lambda_t w_T = -u_{L_T}(c_t, L_t).$$

Replacing these terms in the sum over the budget constraints, the contributions from the wage income parts become:

$$\beta^t \lambda_t (1 - G(A^C, L_{C,t})) w_{M,t} L_{M,t} = -\beta^t u_{L_M}(c_t, L_t) L_M,$$

$$\beta^t \lambda_t w_T L_{T,t} = -\beta^t u_{L_T}(c_t, L_t) L_T.$$

After the telescoping cancellation from asset terms and substituting the expressions for the wage incomes, the summed expression becomes

$$\sum_{t=0}^{\infty} \beta^t \left[\lambda_t c_t - u_{L_M}(c_t, L_t) L_M - u_{L_T}(c_t, L_t) L_T \right] = \lambda_0 R_0 k_0.$$

Recalling that $\lambda_t = u_c(c_t, L_t)$ (and writing $u_c(t)$ for brevity) and $\lambda_0 = u_c(0)$, we obtain

$$\sum_{t=0}^{\infty} \beta^t \left[u_c(t) c_t - u_{L_M}(t) L_M - u_{L_T}(t) L_T \right] = u_c(0) R_0 k_0.$$

Thus, the result. □

This condition ensures that the chosen sequence of consumption, labor, and capital is consistent with the household's optimal choices, initial endowment, and the evolution of prices.

The planner then chooses allocations, in order to maximize the representative household's lifetime welfare subject to (i) the economy's resource constraints, (ii) the implementability condition (8), and (iii) the government's budget constraint.

Formally, the planner's problem is:

$$\max_{\{c_t, k_{t+1}, L_{M,t}, L_{T,t}, L_{C,t}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t} + L_{T,t})$$

subject to

$$c_t + k_{t+1} + \underbrace{F_{L_M}(k_t, k_t^G, L_{M,t})}_{w_{M,t}} L_{C,t} + k_{G,t+1} \leq F(k_t, k_t^G, L_{M,t}) + A^T L_{T,t} + (1 - \delta)k_t + (1 - \delta^G)k_t^G$$

$$k_{G,t+1} = (1 - \delta^G)k_{G,t} + G(A^C, L_{C,t}) \underbrace{F_{L_M}(k_t, k_t^G, L_{M,t})}_{w_{M,t}} L_{M,t},$$

the implementability condition (8) holds.

Proposition 1. *Suppose there exists an interior steady state, then the optimal tax collection choice is determined by:*

$$G_{L_C}(A^C, L_C) = (1 - (1 - \delta_G)\beta) \frac{1}{L_M} \frac{F_k + 1 - \delta}{(F_G - \delta_G) - (F_k - \delta)} \quad (9)$$

Proof. We begin by writing the first-order conditions of the planner's problem for any period $t > 0$.

For the consumption choice we have:

$$\beta^t u_c(t) + \beta^t \mu \left[u_{cc}(t) c_t + u_c(t) - u_{M,cL_M}(t) L_M - u_{T,cL_T}(t) L_T \right] = \lambda_t,$$

where λ_t denotes the Lagrange multiplier on the period- t resource constraint.

The FOC with respect to private capital k_{t+1} is

$$-\lambda_t + \lambda_{t+1} \left[F_k(t+1) + 1 - \delta - F_{L_M} k(t+1) L_{C,t+1} \right] + \gamma_{t+1} G(A^C, L_{C,t+1}) F_{L_M} k(t+1) L_{M,t+1} = 0.$$

Rearranging, we obtain

$$\lambda_t = \lambda_{t+1} \left[F_k(t+1) + 1 - \delta - F_{L_M} k(t+1) L_{C,t+1} \right] + \gamma_{t+1} G(A^C, L_{C,t+1}) F_{L_M} k(t+1) L_{M,t+1}.$$

For the choice of tax collection personnel L_C , the FOC is given by:

$$-\lambda_t w_{M,t} + \gamma_t G_{L_C}(A^C, L_{C,t}) w_{M,t} L_M = 0.$$

Since $w_{M,t} > 0$, it follows that

$$\lambda_t = \gamma_t G_{L_C}(A^C, L_{C,t}) L_M.$$

Finally, the FOC with respect to public capital k_{t+1}^G is

$$-\lambda_t - \gamma_t - \lambda_{t+1} F_{LMG}(t+1) L_{C,t+1} + \left[1 - \delta_G + G(A^C, L_{C,t+1}) F_{LMG}(t+1) L_{M,t+1} \right] \gamma_{t+1} + \lambda_{t+1} \left[F_G(t+1) + 1 - \delta_G \right] = 0.$$

Rearranging this condition yields

$$\lambda_t = \lambda_{t+1} \left[F_G(t+1) + (1 - \delta_G) - F_{LMG}(t+1) L_{C,t+1} \right] - \gamma_t + \left[1 - \delta_G + G(A^C, L_{C,t+1}) F_{LMG}(t+1) L_{M,t+1} \right] \gamma_{t+1}.$$

Dividing by λ_t and writing the resulting expression in terms of ratios, we have

$$1 = \frac{\lambda_{t+1}}{\lambda_t} \left[F_G(t+1) + (1 - \delta_G) - F_{LMG}(t+1) L_{C,t+1} \right] - \frac{\gamma_t}{\lambda_t} + \left[1 - \delta_G + G(A^C, L_{C,t+1}) F_{LMG}(t+1) L_{M,t+1} \right] \frac{\gamma_{t+1}}{\lambda_t}.$$

At an interior steady state the relevant ratios become time-invariant. In particular, the Euler equation implies that $\lambda_{t+1}/\lambda_t = 1/\beta$. Moreover, by setting $L_{C,t} = L_C$ and $L_{M,t} = L_M$ in steady state and using the relation

$$\frac{\gamma}{\lambda} = \frac{1}{G_{LC}(A^C, L_C) L_M},$$

the public capital FOC simplifies to

$$\frac{\gamma}{\lambda} - (1 - \delta_G) \beta \frac{\gamma}{\lambda} = \frac{(F_G - \delta_G) - (F_k - \delta)}{F_k + 1 - \delta}.$$

Substituting $\frac{\gamma}{\lambda} = \frac{1}{G_{LC}(A^C, L_C) L_M}$ into the above expression yields

$$\left[1 - (1 - \delta_G) \beta \right] \frac{1}{G_{LC}(A^C, L_C) L_M} = \frac{(F_G - \delta_G) - (F_k - \delta)}{F_k + 1 - \delta}.$$

Rearranging the final expression, we obtain the desired result:

$$G_{LC}(A^C, L_C) = (1 - (1 - \delta_G) \beta) \frac{1}{L_M} \frac{F_k + 1 - \delta}{(F_G - \delta_G) - (F_k - \delta)}.$$

This completes the proof. □

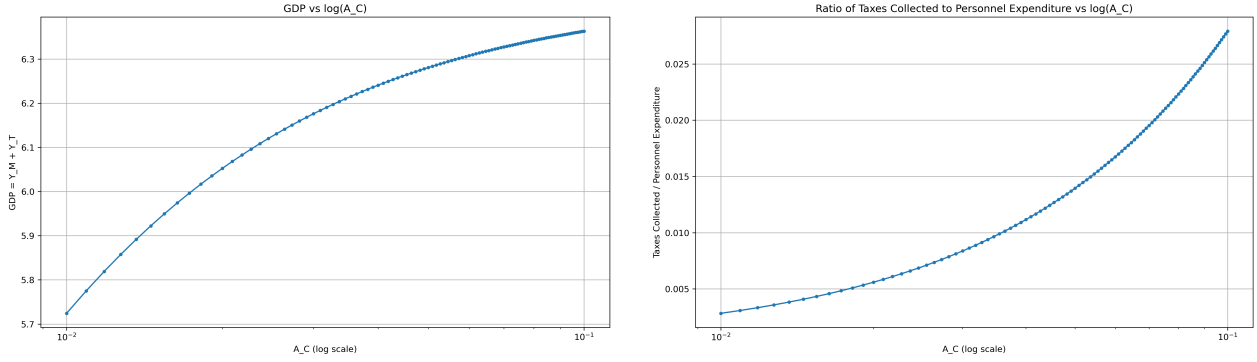
The optimal tax collection rule encapsulates several economic trade-offs. All else equal, a high marginal return on private capital (i.e. a high F_k) makes investment in private capital more attractive relative to public hiring for tax collection. In contrast, a high marginal return on public capital (i.e. a high F_G) means that public capital is particularly productive. This enhanced productivity makes

public hiring more appealing. Finally, improved tax collection productivity (a higher A^C) directly raises the marginal effectiveness of tax collectors, enabling the government to gather higher tax revenues. Finally, a larger modern sector (a higher L_M) expands the taxable base, which naturally increases total tax collections.

Lemma 2. *Suppose that household preferences are separable in consumption and labor, and of type CRRA in consumption, and the tax production function is linear personnel $G(A^C, L_C) = A^C L_C$, then there exists an interior steady state. In addition, the optimal personnel choice over the transition is at each $t > 0$ such that*

$$\frac{(F_G(t+1) - \delta_G) - (F_k(t+1) - \delta)}{F_k(t+1) + 1 - \delta} = \frac{1}{A^C} \left(\frac{1}{L_{M,t}} - \frac{\beta(1 - \delta_G)}{L_{M,t+1}} \right) \quad (10)$$

Figure 3: GDP and Taxes Collected Per Expenditure



4 Quantitative Analysis

A steady is determined by the system of 9 equations and 9 unknowns ($w_M, w_T, r, L_m, L_T, L_C, k, k_G, c$)

$$\text{optimal collection: } G_{L_C}(A^C, L_C) = \frac{1}{L_M} \frac{1}{\beta(F_G + 1 - \delta^G) - 1} \quad (11)$$

$$\text{private capital: } \beta^{-1} = r + (1 - \delta) \quad (12)$$

$$\text{public capital: } k_G = \frac{1}{\delta^G} G(A^C, L_C) w_M L_M \quad (13)$$

$$\text{return on private capital: } r = F_{M,k} \quad (14)$$

$$\text{modern sector wage: } w_M = F_{M,L} \quad (15)$$

$$\text{traditional sector wage: } w_T = F_{T,L} \quad (16)$$

$$\text{modern labor: } u_{L_M} = u_c w_M \times (1 - G(A^C, L_C)) \quad (17)$$

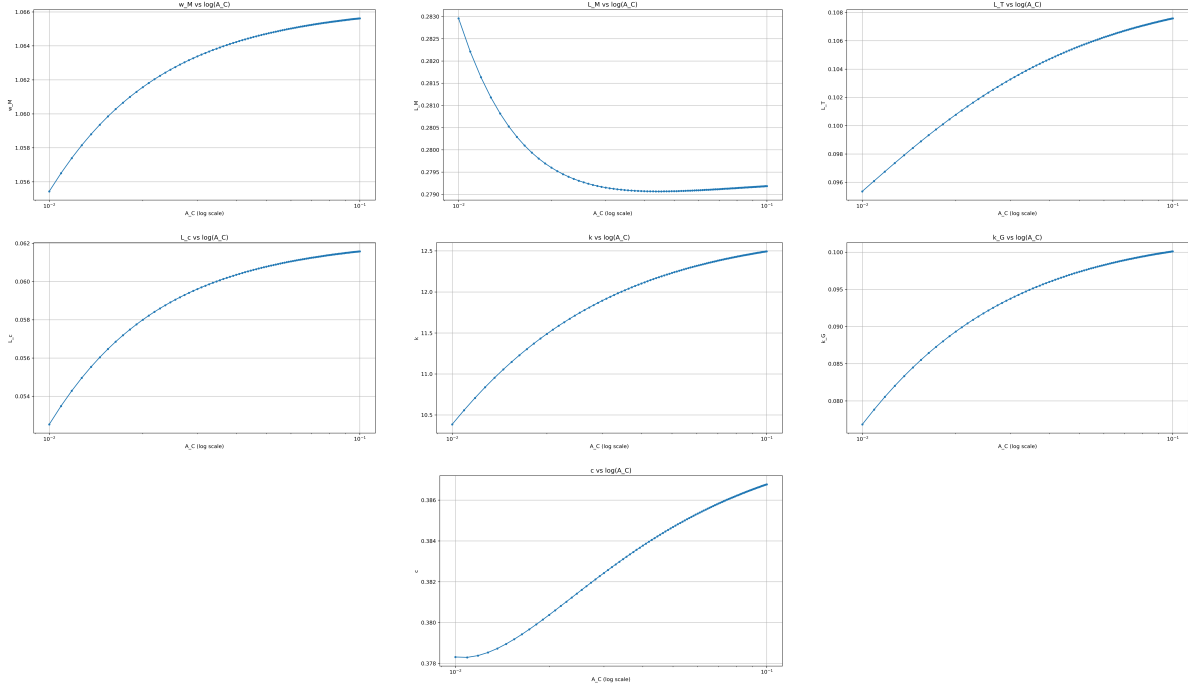
$$\text{traditional labor: } u_{L_T} = u_c w_T \quad (18)$$

$$\text{agg. res. constraint: } c + k + w_M L_C + k_G = (1 + r - \delta)k + w_M L_M + A^T L_T \quad (19)$$

5 Conclusion

This paper develops the concept of tax productivity—the efficiency with which governments convert administrative expenditures into tax revenues—and studies its relationship with economic development. Using comprehensive data covering 98 countries across a wide range of income levels and spanning multiple decades, we document a robust positive association between tax productivity and GDP per capita, observed consistently in both cross-sectional and longitudinal analy-

Figure 4: Steady State Allocations



ses. We find that higher-income countries achieve substantially greater tax productivity, collecting significantly more revenues per dollar spent than lower-income countries. We interpret these empirical patterns through a quantitative macroeconomic model that incorporates tax administration into a Ramsey framework. The model highlights that increased tax productivity enhances a government's ability to finance productive public investments, thus contributing to higher levels of economic output. Preliminary quantitative findings indicate that improvements in tax productivity could substantially impact aggregate outcomes, especially in developing economies.

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